I am an econometrician working on set identification and robust inference. My research asks what can be learned when the information available to the researcher is incomplete: because economically relevant variables are only partially observed, because identifying assumptions are only partially credible, or because economic knowledge does not justify selecting a single model. Rather than resolving these ambiguities through convenient auxiliary assumptions, I develop methods that characterize the conclusions supported by the available information and make explicit which additional conclusions require stronger assumptions.

A central objective of my work is to make this approach operational for empirical researchers. My research therefore combines identification analysis, econometric theory, and computation: I characterize identified sets, develop estimators and inference procedures with formal guarantees, and seek implementations that remain feasible in realistic applications. This perspective also motivates my work in Bayesian econometrics and forecasting, where I develop methods that preserve robustness while remaining computationally tractable.

Job Market Paper

Empirical researchers often work with incomplete datasets in which key covariates or outcome variables are either missing or only interval-valued. Survey designs may reveal variables only up to an interval, administrative data may be top-coded, and data may be missing due to non-random sample attrition. A common applied response is to complete the data before analysis: observations with missing values are either ignored or imputed, observed intervals are replaced by their midpoints, or additional assumptions on the data-generating process are imposed to resolve the incompleteness. All of these strategies can be severely misleading in subsequent regression analysis and may even lead to incorrect conclusions about the sign of a parameter of interest if the underlying assumptions, such as missing at random, are violated.

In my Job Market Paper, *Sharp Identification for Regressions with Interval-Observed and Missing Covariates*, joint with Gil Peled, we ask what the maintained regression model and the observed incomplete data identify on their own. We allow the process generating missingness and interval observations to depend arbitrarily on the latent variables and do not impose missing-at-random, ignorability, or mean-independence restrictions. We derive the sharp identified set, i.e. the smallest set of parameter values consistent with the data and the maintained regression model, together with an associated Hausdorff-consistent estimator and missing- and interval-data-robust confidence intervals for the unknown true parameter.

We study a broad class of conditional moment models with continuous and incompletely observed conditioning variables. Characterizing the sharp identified set in this setting requires establishing whether there exists an infinite-dimensional latent distribution satisfying the infinitely many constraints defined by the conditional moment model. We derive an operational characterization by transforming the original model into an infinite collection of unconditional moments and then extending an existing random-set framework to infinite dimensions. This transforms the problem

into a geometric one and allows us to use inference procedures from the many-moment-inequality literature to construct an estimator and confidence set.

From a practical point of view, our method allows researchers to augment any point estimator based on strong completion assumptions with missing- and interval-data-robust confidence sets. These sets display not only sampling uncertainty but also the ambiguity arising from the incomplete-data problem. In the empirical application, we revisit a multiple-price-list design that elicits discount rates and relates them to on-time high-school graduation. The design itself produces interval-valued covariates, while inconsistent responses and missing outcomes introduce additional sources of incompleteness. Our missing- and interval-data-robust confidence set shows that accounting only for the interval measurement inherent in the design leaves the original qualitative conclusion intact, whereas a non-sharp outer set already includes zero. The conclusion becomes ambiguous only once the additional uncertainty from inconsistent responses and missing outcomes is incorporated.

More broadly, we intend our method to provide a natural robustness benchmark for applied work with incomplete data. If a substantive conclusion survives across all data completions consistent with the observed information and maintained model, it does not depend on a particular imputation rule or auxiliary assumption; if it does not, our method makes that dependence explicit. In this sense, our goal is for the procedure to become a practical robustness tool that applied researchers can report alongside conventional point estimates whenever incomplete data are a concern.

Partial Credibility and Robust Bayesian Inference

In set-identified models, as in my Job Market Paper, the data and maintained assumptions determine a set of admissible parameter values rather than a unique value. From a likelihood perspective, all values within this set are observationally equivalent. A Bayesian researcher may have additional economic knowledge that helps distinguish among them. But this knowledge is often itself incomplete: economic reasoning may tell us, for example, that smaller values of a parameter are more plausible than larger ones without determining a unique probability distribution. Standard Bayesian analysis nevertheless requires a precise joint prior over all structural parameters, forcing the researcher to complete economically meaningful but partial prior information with distributional choices that may have little substantive justification. In my paper *Robust Bayesian Inference under Partially Specified Priors*, I develop posterior inference that uses only the prior information a researcher is willing to defend while remaining robust to how that information is otherwise completed.

This distinction is particularly consequential in set-identified models. Along unidentified dimensions, the likelihood cannot distinguish among observationally equivalent parameter values and therefore does not update the corresponding conditional prior. Consequently, arbitrary features introduced only to obtain a fully specified prior can continue to affect posterior conclusions even

with arbitrarily large samples. I therefore replace a single precise prior with a class of priors that all satisfy the same economically motivated restrictions, for example, restrictions on marginal probabilities, moments, or shape, and characterize the range of posterior conclusions generated by this class.

The structure of set-identified models makes bounds on posterior quantities unusually tractable. Because the likelihood is flat along the unidentified structural dimension, posterior objectives are linear in the unrevisable component of the prior. When economically credible prior information can be expressed through linear restrictions, the robust posterior bounds therefore solve linear programs. I develop a simulation-based approximation that searches over probability weights attached to structural draws, turning an infinite-dimensional robustness problem into a finite-dimensional optimization problem.

From an applied perspective, the framework allows researchers to model partial credibility in identifying assumptions. If the researcher doubts an assumption $\mathcal{A}$, we consider all priors assigning at least $1 - \kappa$ probability to $\mathcal{A}$; varying κ traces a sensitivity frontier between imposing $\mathcal{A}$ with certainty and not imposing it at all. In an oil-market application, I consider prior classes that place most mass on commonly used elasticity bounds. Demand shocks remain more important than supply shocks in explaining real oil-price fluctuations even under substantial weakening of these restrictions, whereas the stronger conclusion that supply shocks are nearly irrelevant does not. More generally, partial prior information offers an alternative to either imposing a single precise prior or allowing full ambiguity: it starts from the economic knowledge one is willing to defend, yielding inference that is more robust than single-prior analysis and less conservative than full-ambiguity robust Bayesian analysis.

Together with my Job Market Paper, this project defines the core of my research agenda. In one case, the incompleteness lies in the data; in the other, it lies in the economic restrictions used for identification. In both cases, I treat the available information as primitive and characterize the full range of conclusions it supports rather than completing that information in an arbitrary way.

Other Research

A second strand of my research develops econometric methods that add flexibility while remaining easy to implement. In joint work with Massimiliano Marcellino and Tommaso Tornese, *Power Prior for VARs*, we introduce the power-prior framework to Bayesian VARs as a simple way to borrow information from external datasets. External likelihoods enter with estimated discount factors, so borrowing adapts to the relevance of each donor. We extend the theory to VARs, preserve conjugacy and compatibility with standard structural-identification methods, and show in simulations and applications that cross-country borrowing can improve forecasts and tighten inference without estimating a full multicountry system.

Two further projects emphasize implementation. In *Conformal Quantile Estimation in Economics*, I

use one-sided conformal prediction to turn generic point forecasts into calibrated quantile forecasts with low computational cost and apply this procedure to nowcasting GDP. In joint work with Jakub Przewoski and Gabriele Nespoli, *Model Confidence Set for Stata*, we provide a Stata implementation of the model confidence set procedure. Across these projects, I view computational tractability and ease of use as an integral part of the econometric contribution.

Future Research

In future work, I plan to build further on my research agenda in set identification and robust inference, particularly in settings where commonly used econometric objects are themselves only partially observed. One direction concerns rank-based methods when the variables used to construct ranks are incomplete. For example, missing income observations make not only the ranks of the missing individuals but also those of observed individuals jointly uncertain. I am interested in identification and inference for rank-rank regressions in such settings. The discrete ordering structure of ranks provides additional information that is absent in generic interval-data problems and may allow for particularly tractable sharp characterizations. This project could for instance be used to study rank-rank regression if the underlying variable is missing.

A related direction concerns the connection between moment-inequality inference and inference on optimization problems. Many partially identified models admit both representations: feasibility can be expressed through a collection of inequalities or through the value of an associated optimization problem, with near-binding moments corresponding to active constraints. I am interested in understanding when moment-selection and value-function-based methods exploit the same local geometry, and whether this connection can lead to simpler or sharper inference in high- and infinite-dimensional settings.

I also plan to continue my work in macroeconometrics. One question I am interested in is how to interpret uncertainty reported in probabilistic surveys of inflation expectations. A dispersed subjective forecast may reflect uncertainty about future inflation itself, but also uncertainty about the respondent's own information or model of the economy. I am interested in developing survey and econometric methods that separate these sources of uncertainty and studying whether they respond differently to information. More broadly, I believe macroeconomics offers many important settings in which model misspecification and partial identification remain underexplored, and I am interested in bringing the tools developed in my other work to these questions.